

4	(a)	The <u>maximum displacement</u> of the mass from the equilibrium position/fixed position. [B1]	
	(b)	(i)	Spring constant $k = \frac{F}{e}$ $= \frac{mg}{e} = \frac{0.25 \times 9.81}{0.050}$ [C1] $= 49.1 \text{ N m}^{-1}$ [A1]



		(ii)	work done = gain in elastic P.E. - loss in GPE $= \frac{1}{2}k(e_2^2 - e_1^2) - mgh$ $= \frac{1}{2}(49.05)(0.100^2 - 0.050^2) - 0.25(9.81)(0.050) \quad [M1]$ $= 0.062 \text{ J} \quad [A1]$	
		(iii)	total energy = work done = 0.062J [A1]	
		(iv)	$\omega = \frac{2\pi}{T} = \frac{2\pi}{2\pi\sqrt{\frac{m}{k}}} = \sqrt{\frac{k}{m}} = \sqrt{\frac{49.05}{0.25}} = 14.0 \text{ rad s}^{-1} \quad [C1]$ $v = \pm\omega\sqrt{x_0^2 - x^2} = \pm(14.0)\sqrt{(0.050)^2 - (0.020)^2} \quad [C1]$ $= \pm 0.642 \text{ m s}^{-1} \quad [A1]$	

5	(a)	(i)	No net heat flow/transfer between the objects. [B1]
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